1 Problem setup

$$\dot{x} = -0.1\,yz, \quad \dot{y} = 1.6\,xz, \quad \dot{z} = -0.75\,xy$$

$$u_0 = (0.5,\ 0.75,\ 1.0), \qquad t \in [0,\ 13.26446]$$

27 windows · 1,536 points/window · float64

REPEAT FOR EACH WINDOW $k = 0 \ldots 26$

2 PINN for window k

Input t → **Hidden** 4×60 tanh → **Output** $N_k(t) \in \mathbb{R}^3$ → **Hard IC** $\hat{u}_k = u_{0,k} + (t - t_k)\,N_k(t)$

3 ODE residual

$r_k = d\hat{u}_k/dt - f(\hat{u}_k)$; derivatives by autodiff

4 Causal residual loss

$$\mathcal{L}_i = \mathrm{mean}(r_i^2), \quad w_i = \exp\left(-\varepsilon \sum_{j<i} \mathcal{L}_j\right)$$

$\mathcal{L} = \mathrm{mean}(w_i \mathcal{L}_i)$; detach weights for gradients

$\varepsilon :\ 0.01 \to 0.1 \to 1 \to 10$

5 Adam optimization

learning rate × 0.9 every 1,000 steps

Stage complete? $\min w > 0.99$ or 4,000 steps

NO → back to 4 Causal residual loss

YES → **All ε values used?**

NO → **Next ε** keep parameters → next ε → back to 4 Causal residual loss

YES →

6 L-BFGS

≤ 1,000 iterations

unweighted residual MSE

7 Save window

checkpoint · history · prediction

Last window? $k = 26$

NO → **Next window** $u_{0,k+1} = \hat{u}_k(t_{k+1})$ copy parameters $k \leftarrow k + 1$ → next window → back to 2 PINN for window k

YES →

8 Reference evaluation

DOP853 · 13,265 points · rtol 10^{-10} · atol 10^{-12}

evaluation only; not used for optimization

9 Evaluation outputs

MAE · RMSE · maximum error · residual MSE

invariant drift · training cost